Worksheet 5-1 A MVT #1
AP Calculus

Name SOLUTIONS

Directions: Solve each problem.

1.) Skater Sully is riding a skateboard back and forth on a street that runs north/south. The twice-differentiable function S models Sully's position on the street, measured by how many meters north he is from his starting point, at time t, measured in seconds from the start of his ride. The table below gives values of $S(t)$ at selected times t.

t seconds	0	20	30	60
$S(t)$ meters	0	-5	7	40

a.) For $0 \le t \le 20$, must there be a time t when Sully is 2 meters south of his starting point? Justify your answer.

S is a twice-differentiable function so S is a continuous function.
$S(t) = -2$ is between $S(0) = 0$ and $S(20) = -5$
So the IVT guarantees there must be a time t in $0 \le t \le 20$ such that $S(t) = -2$

b.) For $30 \le t \le 60$, must there be a time t when Sully's velocity is 1.1 meters per second? Justify your answer.

S is a twice-differentiable function so $v(t) = S'(t)$ is a continuous function.
$\frac{S(60)-S(30)}{60-30} = \frac{40-7}{30} = \frac{33}{30} = 1.1$
So the MVT guarantees there must be a time t in $30 \le t \le 60$ such that
$v(t) = S'(t) = \frac{S(60)-S(30)}{60-30} = 1.1$ meters per second.

2.) A particle is moving along the x-axis. The twice-differentiable function s models the particles distance from the origin, measured in centimeters, at time t, measured in seconds.

t seconds	3	10	20	25
$s(t)$ cm	5	-2	-10	8

a.) For $20 \le t \le 25$, must there be a time t when the particle is at the origin? Justify your answer.

s is a twice-differentiable function so s is a continuous function.
$s(t) = 0$ is between $S(20) = -10$ and $S(25) = 8$
So the IVT guarantees there must be a time t in $20 \le t \le 25$ such that $s(t) = 0$

b.) For $3 \le t \le 10$, must there be a time t when the particle's velocity is –1 cm per second? Justify your answer.

s is a twice-differentiable function so $v(t) = s'(t)$ is a continuous function.
$\frac{s(10)-s(3)}{10-3} = \frac{-2-5}{7} = \frac{-7}{7} = -1$
So the MVT guarantees there must be a time t in $3 \le t \le 10$ such that
$v(t) = s'(t) = \frac{s(10)-s(3)}{10-3} = 1$ cm per second.

Directions: Select the best answer for each question. Be sure to show your work!

2009 Practice Exam #86 (AB)

B 3.)

x	3	4	5	6	7
$f(x)$	20	17	12	16	20

The function f is continuous and differentiable on the closed interval [3, 7]. The table above gives selected values of f on this interval. Which of the following statements must be true?

I. The minimum value of f on [3, 7] is 12.
II. There exists c, for $3 < c < 7$, such that $f'(c) = 0$.
III. $f'(x) > 0$ for $5 < x < 7$

(A) I only
(B) I only
(C) III only
(D) I and III only
(E) I, II, and III

I. The minimum could be 12
II. By the MVT, there is a c for 3 < c < 7, such that $f'(c) = \frac{f(7)-f(3)}{7-3} = \frac{20-20}{4} = 0$
III. We only know the function is continuous, we don't know if it is strictly increasing, so we can's say for sure $f'(x) > 0$ for $5 < x < 7$

1998 #91 (AB)

E 4.) Let f be a function that is differentiable on the open interval $(1, 10)$. If $f(2) = -5$, $f(5) = 5$, and $f(9) = -5$, which of the following must be true?

I. f has at least 2 zeros.
II. The graph of f has at least one horizontal tangent
III. For some c, $2 < c < 5$, $f(c) = 3$

This means $f'(x) = 0$

(A) None
(B) I only
(C) I and II only
(D) I and III only
(E) I, II, and III

I. True. The IVT guarantees $f(x) = 0$ in [2, 5] and [5, 9]
II. True. The MVT guarantees there is a c for 2 < c < 9, such that
$f'(c) = \frac{f(9)-f(2)}{9-2} = \frac{-5-(-5)}{7} = 0$
III. True. The IVT guarantees $f(x) = 3$ in [2, 5]

1998 #26 (AB)

A 5.)

x	0	1	2
$f(x)$	1	k	2

The function f is continuous on the closed interval [0, 2] and has values that are given in the table above. The equation $f(x) = \frac{1}{2}$ must have at least two solutions in the interval [0. 2] if $k =$

(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2 (E) 3

To guarantee 2 solutions in the interval by the IVT, $k < \frac{1}{2}$

2012 #77 (AB)

A 6.) Let f be a function that is continuous on the closed interval $[2, 4]$ with $f(2) = 10$ and $f(4) = 20$. Which of the following is guaranteed by the Intermediate Value Theorem?

(A) $f(x) = 13$ has at least one solution in the open interval $(2, 4)$.
(B) $f(3) = 15$
(C) f attains a maximum on the open interval $(2, 4)$.
(D) $f'(x) = 5$ has at least one solution in the open interval $(2, 4)$.
(E) $f'(x) > 0$ for all x in the open interval $(2, 4)$.

(B) IVT guarantees a value where $f(c) = 15$ with $2 < c < 4$, but it doesn't have to be $c = 3$
(C) IVT does not guarantee anything about a maximum value
(D) MVT would guarantee $f'(c) = \frac{f(4)-f(2)}{4-2} = \frac{30-1}{2} = 10$ in $2 < c < 4$
(E) neither IVT or MVT guarantee the value for $f'(x)$ for all x in the interval

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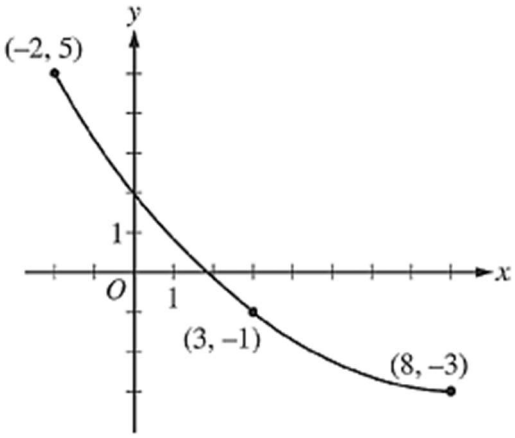
2013 #85 (BC)

C 7.) A portion of the graph of a differentiable function f is shown at the right. If the value $c = 3$ satisfies the conclusion of the Mean Value Theorem applied to f on the open interval $-2 < x < 8$, what is the slope of the line tangent to the graph of f at $x = 3$?

(A) $-\frac{7}{5}$ (B) $-\frac{5}{4}$ (C) $-\frac{4}{5}$
(D) $-\frac{5}{7}$ (E) $-\frac{1}{5}$



Graph of f

Slope of the tangent to the graph of f at $x = 3$ is $f'(3)$. Since $c = 3$ satisfies the conclusion of the Mean Value Theorem on $-2 < x < 8$, we know $f'(3) = \frac{f(8)-f(-2)}{8-(-2)} = \frac{-3-5}{10} = \frac{-8}{10} = \frac{-4}{5}$

1985 #13 (BC)

B 7.) Let f be the function given by $f(x) = x^3 - 3x^2$. What are all values of c that satisfy the conclusion of the Mean Value Theorem of differential calculus on the closed interval $[0, 3]$?

(A) 0 only (B) 2 only (C) 3 only (D) 0 and 3 (E) 2 and 3

$f(0) = 0 - 0 = 0$ and $f(3) = 27 - 27 = 0$ $\quad f'(x) = 3x^2 - 6x$
By the MVT, $f'(c) = \frac{f(3)-f(0)}{3-0} = \frac{0-0}{3} = 0$ for some c in $(0, 3)$
$3c^2 - 6c = 0$
$3c(c - 2) = 0$
$c = 0$ and $c = 2$ 2 is the only value in $(0, 3)$